

EduMiners – Datathon 2026 Report

Diagnosing Foundational Mathematics Gaps Across Rural Karnataka

1. Summary

We analysed **1.38 million student assessment records** from the Akshara GP Maths Contest (Grades 4–6, 2022–25) across 31 Karnataka districts. The contest data was integrated with **NFHS-5** socio-economic indicators and **2025 SSLC Mathematics** results to understand factors influencing foundational mathematics performance.

Our analysis found that district-level mathematics performance is strongly associated with development indicators such as female schooling and sanitation. We also observed that districts performing poorly in Grades 4–6 tend to perform poorly in the Class-10 Mathematics Board examination (Spearman $\rho = 0.57$). A diagnostic regression model ($R^2 = 0.43$) was used to identify districts performing below their development-based expectations, helping distinguish where school-level interventions may have the greatest impact.

2. Dataset & Methodology

Datasets

- **Akshara GP Maths Contest** (~1.38M student records, Grades 4–6, 2022–25)
- **NFHS-5** district-level development indicators
- **SSLC 2025 Mathematics** district pass percentages

Methodology

Our reproducible Python pipeline performs:

- Student score calculation
- District-level aggregation
- Competency analysis
- NFHS correlation analysis
- SSLC correlation analysis
- Development Index construction
- OLS regression for district benchmarking
- Residual-gap analysis for intervention prioritisation

3. Key Findings

District disparities dominate

Overall mathematics accuracy remains close to **53%**, with much larger variation **between districts** than between grades, indicating that geography is a stronger predictor of learning outcomes than grade progression.

Development strongly influences learning

Districts with higher levels of female schooling, women's education and sanitation consistently achieve better mathematics performance, while indicators such as child marriage and malnutrition show negative associations.

Learning gaps persist to secondary education

Primary-school mathematics performance correlates with **SSLC Mathematics pass percentage ($p = 0.57$)** across districts, suggesting that foundational learning gaps continue through later schooling.

Diagnostic modelling

The Development Index explains **43%** of district-level variation in mathematics accuracy. Residual-gap analysis identifies districts such as **Chitradurga, Davanagere and Ballari** that perform below their expected level, making them strong candidates for targeted school-level interventions.

4. Recommendations

- Prioritise districts with large negative residual gaps for academic interventions.
- Focus remediation on weaker competencies such as division and multiplication.
- Continue annual monitoring using the reproducible pipeline.
- Integrate educational planning with broader development initiatives including sanitation and women's education.